

FIM 590 Algo. and HFT

Homework #2

Due: October 1st, 2026

1. **Problem E.6.1 from C-J-P (2015).** An agent wishes to liquidate $\mathfrak{N}$ shares between t and T using market orders. The value function is

$$H(t, s, q) = \sup_{\nu \in \mathcal{A}_{t,T}} \mathbb{E}_{t,s,q} \left[\int_t^T (S_u - k\nu_u) \nu_u du + Q_T^\nu (S_T - \alpha Q_T^\nu) \right],$$

where $k > 0$ is the temporary market impact, ν_t is the speed of trading, $\alpha \geq 0$ is the liquidation penalty, and $dS_t = \sigma dW_t$.

- (a) Show that the value function H satisfies

$$0 = - \left(\frac{\partial}{\partial q} H - s \right)^2 - 4k \frac{\partial}{\partial t} H - 2k\sigma^2 \frac{\partial^2}{\partial s^2} H.$$

- (b) Make the ansatz

$$H(t, s, q) = h_2(t)q^2 + h_1(t)q + h_0(t) + qs$$

and show that the optimal liquidation rate is

$$\nu_t^* = \frac{Q_t^{\nu^*}}{T - t + \frac{k}{\alpha}}. \quad (1)$$

- (c) Let $\alpha \rightarrow \infty$ and show that (1) converges to the TWAP strategy. Moreover, discuss the intuition of the strategy when $\alpha \rightarrow 0$.

2. **Problem E.6.2 from C-J-P (2015).** This exercise is similar to that above but with a slightly different setup. The agent wishes to liquidate $\mathfrak{N}$ shares and her objective is to maximize expected terminal wealth which is denoted by X_T^ν (in the exercise above we wrote terminal wealth as $\int_t^T (S_u - k\nu_u) \nu_u du$). The value function is

$$H(t, x, s, q) = \sup_{\nu} \mathbb{E}_{t,x,s,q} [X_T^\nu + Q_T^\nu (S_T - \alpha Q_T^\nu)], \quad (6.41)$$

where

$$dX_t^\nu = (S_t - k\nu_t) \nu_t dt. \quad (6.42)$$

- (a) Show that the HJB satisfied by the value function $H(t, s, q, x)$ is

$$0 = \left(\frac{\partial}{\partial t} + \frac{1}{2} \sigma^2 \frac{\partial^2}{\partial s^2} \right) H + \sup_{\nu} \left\{ \left(-\nu \frac{\partial}{\partial q} + (s - k\nu) \nu \frac{\partial}{\partial x} \right) H \right\}, \quad (6.43)$$

and the optimal liquidation rate in feedback form is

$$\nu_t^* = \frac{\frac{\partial}{\partial q} H - s \frac{\partial}{\partial x} H}{-2k \frac{\partial}{\partial x} H}. \quad (6.44)$$

- (b) To solve (6.43), use the terminal condition $H(T, x, s, q) = x + qs - \alpha q^2$ to propose the ansatz

$$H(t, s, x, q) = x + h(t)q^2 + qs, \quad (6.45)$$

where $h(t)$ is a deterministic function of time. Show that

$$h(t) = -\frac{k}{T - t + \frac{k}{\alpha}}, \quad (6.46)$$

and

$$\nu_t^* = \frac{Q_t^{\nu^*}}{T - t + \frac{k}{\alpha}}.$$

3. **(Running Inventory Risk).** An agent liquidates $\mathfrak{N}$ shares over $[0, T]$ subject to temporary price impact parameter $k > 0$ and running inventory risk penalty parameter $\phi > 0$. The value function is

$$H(t, s, q) = \sup_{\nu \in \mathcal{A}} \mathbb{E}_{t,s,q} \left[\int_t^T ((S_u - k\nu_u)\nu_u - \phi Q_u^2) du \right],$$

subject to $dQ_t = -\nu_t dt$, $dS_t = \sigma dW_t$, and hard terminal constraint $Q_T = 0$.

- (a) Make the additive ansatz $H(t, s, q) = qs + h(t, q)$ with quadratic penalty $h(t, q) = -h_2(t)q^2$. Show that $h_2(t)$ satisfies

$$\dot{h}_2(t) - \frac{1}{k}h_2^2(t) + \phi = 0.$$

- (b) Enforce the terminal condition $\lim_{t \rightarrow T} h_2(t) = \infty$ (corresponding to $Q_T = 0$) to show that

$$h_2(t) = \sqrt{k\phi} \coth \left(\sqrt{\frac{\phi}{k}}(T - t) \right).$$

- (c) Show that the optimal inventory trajectory is given by

$$Q_t^* = \mathfrak{N} \frac{\sinh(\gamma(T - t))}{\sinh(\gamma T)}, \quad \text{where } \gamma = \sqrt{\frac{\phi}{k}},$$

and verify that as $\phi \rightarrow 0$ (risk-neutral limit), Q_t^* converges back to the standard linear TWAP trajectory $Q_t^* = \mathfrak{N} \left(1 - \frac{t}{T}\right)$.